Sofía Cominelli

Ph.D. Candidate · Soil Fertility and Nutrient Management
Department of Agronomy, Kansas State University
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Education

January 2024 to present | Kansas State University | Manhattan, KS

Doctor of Philosophy in Agronomy (Anticipated Graduation Dec 2026)

January 2022 to December 2023 | Kansas State University | Manhattan, KS

Master of Science Agronomy

May 2019 | National University of Rosario | Rosario, Argentina

Agronomic Engineer

Professional Experience

2024 – present: Graduate Research Associate in Soil Fertility and Nutrient Management, Dept. of Agronomy, Kansas State University: Initiate and perform experiments in the field and laboratory relevant to the completion of doctorate dissertation.

2022-2023: Graduate Research Assistant, Soil Water Process, Dept. of Agronomy, Kansas State University: Initiate and perform field, greenhouse and laboratory experiments relevant to my master's thesis, provide assistance to other graduate students in the soil water process research program and department regarding, implementation in the lab, greenhouse or field.

2021: Visiting Scholar, Soil Water Process, Dept. of Agronomy, Kansas State University. Initiate and perform field, greenhouse and laboratory experiments relevant to the soil water program, developed a greenhouse experiment to assess Palmer amaranth transpiration response to drying soil.

2020-2021: Service and technological implementations GeoAgro by TEK: development of planting, harvesting, fertilization, and herbicide application maps, and training and assisting clients.

2019 Supported the organization and on-site execution during the 27th AAPRESID Congresses, coordinating conference rooms, welcoming and assisting speakers, and ensuring the smooth delivery of scheduled sessions.

2018 - 2019: Undergraduate internship in National Institute of Agricultural Technology: Assist with planting, harvesting and data collection in soybean, vetch and alfalfa.

2017-2019: Undergraduate Research Assistant, Weed Ecology Team, College of Agriculture, National University of Rosario: Assist with planting, weed control, data collection and preparing in-class demonstrations.

2017- 2018 Undergrad Research Group on Management and Utilization of Extensive Crops: Assist with planting, harvesting, fertilization and data collection in sorghum.

2017 Supported the organization and on-site execution during the 25th AAPRESID Congresses, coordinating conference rooms, welcoming and assisting speakers, and ensuring the smooth delivery of scheduled sessions.

Research And Publications

Peer-Reviewed Journal Papers

Baral, R., Kim, J., Bhattarai, B., Koirala, H., Massigoge, I., Denson, E., Cominelli, S ... & Min, D. (2025). Cropping potential of forage soybean as a summer forage in Midwest US rainfed systems. Frontiers in Agronomy, 7, 1570567.

Cominelli, S., Rivera, L. D., Brown, W. G., Ochsner, T. E., & Patrignani, A. (2024). Calibration of TEROS 10 and TEROS 12 electromagnetic soil moisture sensors. Soil Science Society of America Journal, 88(6), 2104-2122.

Massigoge, I., Baral, R., Cominelli, S., Denson, E., Helguera, P. G., Guareschi, C., ... & Ciampitti, I. A. (2024). Exploring alternative crop rotations to continuous winter wheat for agricultural intensification in the US central Great Plains. Agricultural Systems, 216, 103879. https://doi.org/10.1016/j.agsy.2024.103879

Cominelli, S., & Patrignani, A. (2022). Transpiration response of palmer amaranth (Amaranthus palmeri) to drying soil in greenhouse conditions. Frontiers in Agronomy, 4, 1018251. https://doi.org/10.3389/fagro.2022.1018251

Patrignani, A., Parker, N., & Cominelli, S. (2022). Upland Rootzone Soil Water Deficit Regulates Streamflow in a Catchment Dominated by North American Tallgrass Prairie. Water, 14(5), 759.

Under Review Journal Papers

Cominelli, S., Lacasa, J., Cott, R. L., Hettiarachchi, G. M., & Ruiz Diaz, D. A. (2026). Comparison of Statistical Methods to Characterize Yield Response to Phosphorus. European Agronomy Journal. Code available: https://github.com/SCominelli/Comparison-of-Statistical-Methods-to-Characterize-Yield-Response-to-Phosphorus-Simulation-Code

Abstracts Presented

Cominelli, S., Hettiarachchi, G. M., Cott, R. L., Lacasa, J., & Ruiz Diaz, D. A. (2025) Evaluating Near-Infrared Models for Predicting Soil Chemical Properties in Kansas. CANVAS 2025, Salt Lake City, UT.

Cominelli, S., & Ruiz Diaz, D. A. (2024) NIR-Based Estimations of Soil Parameters. ASA, CSSA, SSSA International Annual Meeting, San Antonio, TX. https://scisoc.confex.com/scisoc/2024am/meetingapp.cgi/Paper/160540

Baral, R., Kim, J., Bhattarai, B., Denson, E., Guareschi, C. A., Cominelli, S., Peraza, J., Bezerra de Oliveira, J., Garcia Helguera, M. P., Ciampitti, I. A., Rice, C. W., Massigoge, I., & Min, D. (2024) Cropping Potential of Forage Soybean in the Midwest Rainfed System [Abstract]. ASA, CSSA, SSSA International Annual Meeting, San Antonio, TX. https://scisoc.confex.com/scisoc/2024am/meetingapp.cgi/Paper/160500

Guareschi, C. A., Mendes, A. C., Massigoge, I., Baral, R., Garcia Helguera, M. P., Cominelli, S., Peraza, J., Simão, L. M., Widanagamage, N., Denson, E., Parajuli, S., Patrignani, A., Lollato, R. P., Min, D., Ruiz Diaz, D. A., Ciampitti, I. A., Santos, E. A., & Rice, C. W. (2024) The Effect of Crop Intensification and Diversification on Soil Health Under Rainfed Wheat-Based Cropping Systems [Abstract]. ASA, CSSA, SSSA International Annual Meeting, San Antonio, TX. https://scisoc.confex.com/scisoc/2024am/meetingapp.cgi/Paper/158835

Cominelli, S., Lollato, R. P., Simão, L. M., & Patrignani, A. (2023) Quantifying in Situ soil Evaporation Using a Model-Data Assimilation Approach [Abstract]. ASA, CSSA, SSSA International Annual Meeting, St. Louis, MO. https://scisoc.confex.com/scisoc/2023am/meetingapp.cgi/Paper/150908

Cominelli, S., & Patrignani, A. (2023) Automating Soil Evaporation Measurements Using Instrumented Microlysimeters. [Abstract]. ASA, CSSA, SSSA International Annual Meeting, St. Louis, MO. https://scisoc.confex.com/scisoc/2023am/meetingapp.cgi/Paper/150968

Patrignani, A., Ochsner, T. E., Wyatt, B. M., Cosh, M., Brown, W. J., Azizi, S., Peraza, J., Cominelli, S., Widanagamage, N., & Parker, N. (2023) New in-Situ Testbeds for Improving Soil Moisture Sensing [Abstract]. ASA, CSSA, SSSA International Annual Meeting, St. Louis, MO. https://scisoc.confex.com/scisoc/2023am/meetingapp.cgi/Paper/152761

Massigoge, I., Baral, R., Cominelli, S., Denson, E., Garcia Helguera, M. P., Guareschi, C. A., Simão, L. M., Peraza, J., Bonini Pires, C., Dille, J., Lollato, R. P., Min, D., Patrignani, A., Ruiz Diaz, D. A., Hefley, T., Lira, S., Prasad, P. V. V., Rice, C. W., Hatfield, J. L., & Ciampitti, I. A. (2023) Investigating Alternative Crop Rotations for Agricultural Intensification in the US Central Great Plains [Abstract]. ASA, CSSA, SSSA International Annual Meeting, St. Louis, MO. https://scisoc.confex.com/scisoc/2023am/meetingapp.cgi/Paper/148585

Cominelli, S., Lollato, R. P., Simão, L. M., & Patrignani, A. (2022) Evapotranspiration Partitioning of Rainfed Winter Wheat [Abstract]. ASA, CSSA, SSSA International Annual Meeting, Baltimore, MD. https://scisoc.confex.com/scisoc/2022am/meetingapp.cgi/Paper/143295

Simão, L. M., Patrignani, A., Cominelli, S., & Lollato, R. P. (2022) Adjusting Sowing Date to Reduce the Effect of Elevated Temperature in Winter Wheat [Abstract]. ASA, CSSA, SSSA International Annual Meeting, Baltimore, MD. https://scisoc.confex.com/scisoc/2022am/meetingapp.cgi/Paper/141850

Extension Publications

Roa, G. A., Rutter, E. B., Sohm, C. D., Cominelli, S., Demarco, J., Sivarathri, B. S., & Sen, S. (2026). Understanding the foundations of fertilizer recommendations. CSA News.

Xin, X., Nepal, J., Bezerra de Oliveira, J., Roa Acosta, G., Cominelli, S., & Maryam, H. (2025). Managing soil N for productivity and sustainability: Understanding gains and losses. CSA News.

Ghimire, O. P., Cominelli, S., Bezerra de Oliveira, J., Sivarathri, B. S., Seelam, R. T., Kaushal, S., Khanal, B., & Joshi, P. (2025). Agronomic approaches for enhancing nutritional value in food crops. CSA News.

Simão, L. M.; Patrignani, A.; Cominelli, S.; and Lollato, R. P. (2023) "Previous Crop Impacts Winter Wheat Sowing Dates, Available Water at Sowing, and Grain Yield," Kansas Agricultural Experiment Station Research Reports: Vol. 9: Iss. 4. https://doi.org/10.4148/2378-5977.8473

Under Review Extension Publications

Cominelli, S., Ruiz Diaz D.A. (2026) "Near-infrared Spectroscopy for Predicting Soil Chemical Properties in Kansas", Kansas Agricultural Experiment Station Research Reports.

Cominelli, S., Ruiz Diaz D.A. (2026)" Evaluating Soil Test Phosphorus Categories for Kansas Winter Wheat ", Kansas Agricultural Experiment Station Research Reports.

Roa, G. A., Cominelli, S., Demarco, J.O., Ruiz Diaz, D.A. (2026) “Corn Grain Yield Response to Phosphorus Rate, Source, and Application Timing in Kansas”, Kansas Agricultural Experiment Station Research Reports.

Roa, G. A., Demarco, J.O., Cominelli, S., Ruiz Diaz, D.A. (2026) “Nitrogen Fertilization Effects on Corn, Wheat, and Soybean under Crop Rotations in Kansas”, Kansas Agricultural Experiment Station Research Reports.

Research Funding

2026 | Principal investigator “Soil and weather predictors affecting the probability of response to Phosphorus” OCP North America. Award amount: $16,000. Co-investigator: Dorivar Ruiz Diaz, Josefina Lacasa, Brain Bohman, and Jagmandeep Dhillon.

Teaching

AGRON 385 – Soil Analysis and Nutrient Recommendations, Spring 2025

AGRON 220 – Crop Science Lab, Kansas State University, Fall 2024

AGRON 745 – Environmental Soil Physics, Kansas State University, Spring 2023

Leadership, awards, scholarships and community involvement

Leadership and Service

Research Hour Collaborative Initiative, Ruiz Diaz Research Group, Kansas State University

Founder and Facilitator | Spring 2025 to present

Created a weekly peer research and team building initiative for a nine-student group (paper writing, abstracts, posters, presentations, journal discussions, troubleshooting, and statistical analysis).

Outcomes (Spring 2025 to Fall 2025): team won First Place in Hackathon (Nitrogen Use Efficiency Workshop), earned three poster competition awards, and submitted and presented 20 abstracts.

International Coordinating Council (ICC), Kansas State University

Webmaster Officer | Spring 2024 to Spring 2025

Led communications and digital engagement; grew ICC Instagram from approximately 300 to about 700 followers.

Founded and hosted the ICC K-State Podcast to elevate international voices and strengthen cross cultural understanding.

Hosted panel discussion: “Thriving Abroad: Stories of Connection and Resilience.”

Supported major events: Parade of Flags, Game Nights, Food Festival, International Championship, International Week.

Grad Edge Program, Staley School of Leadership, Kansas State University | Spring 2024

Completed leadership development training and applied lessons to create and facilitate the Research Hour Collaborative Initiative in Spring 2025.

Agronomy Graduate Student Association (AGSA), Kansas State University

Apparel Committee Chair | 2024

Coordinated AGSA apparel order and distribution, delivering 70 shirts to strengthen departmental pride and belonging among international graduate students.

Professional Development Committee Chair | 2024

Organized a “Next Steps After Graduation” Q&A with International Student and Scholar Services, a Women’s Mentorship Breakfast, and an Elevator Pitch and Networking Night with 60+ attendees.

Social Media Officer / Chair (elected) | 2023

Managed Instagram, Facebook, LinkedIn, and YouTube; created “Kudos Corner”; grew AGSA Instagram from approximately 100 to nearly 400 followers.

Scholarship and Awards

2026 | Morehouse scholarship

2025 | Ray Lamond Memorial Graduate Scholarship in Agronomy

2025 | 55th NCSFC Graduate Student Award

2025 | First Place in Hackaton, Nitrogen Use Efficiency Workshop

2024 | Ernest L. and Lou R. Adams Scholarship Fund in Agriculture

2023 | Irvin D. and Dora Mae Atkins Family Scholarship

2023 | First Oral Student Competition, Evapotranspiraton Measurement and Modeling community, ASA

2023 | Agronomy Graduate Student Association Scholarship

2022 | Schrader/Massier Graduate Excellence Fund in Agronomy

Organization and Community Involvement

2022 - Present | Agronomy Graduate Student Association | Kansas State University

2022- Present | American Society Agronomy (ASA), Soil Science Society of America (SSSA), and Crop Science Society of America (CSSA)

2024 - 2025 | Soil Water Conservation Chapter | Kansas State University

2024 - 2025 | International Coordinating Council | Kansas State University

2019-2021 | Argentinian Female Agronomist

2017 - 2021 | AAPRESID Joven